Year 10 Mathematics

Shape and Space 1

5.3 Pythagoras' Theorem

Pythagoras' Theorem

STARTER · 5 MIN

Date: _____ **Title:** Pythagoras' Theorem

LEARNING OBJECTIVES

- Find the hypotenuse of a right-angled triangle
- Find a shorter side of a right-angled triangle
- Use Pythagoras' theorem to solve problems

STARTER *Answer in your book. No calculators.*

1. Work out 6^2
2. Work out $\sqrt{81}$
3. Work out $3^2 + 4^2$
4. Work out $13^2 - 5^2$

Starter — Answers

STARTER

Tick or correct in green pen

1. $6^2 = \mathbf{36}$

2. $\sqrt{81} = \mathbf{9}$

3. $9 + 16 = \mathbf{25}$

4. $169 - 25 = \mathbf{144}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the hypotenuse of a right-angled triangle	☐
SILVER	I can find a shorter side of a right-angled triangle	☐
GOLD	I can use Pythagoras' theorem to solve problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Pythagoras' Theorem

INTRODUCTION · 3 MIN

In a right-angled triangle: $a^2 + b^2 = c^2$

Hypotenuse

The longest side, opposite the right angle.

This is c

Find c

Square, add, then square root.

$$c = \sqrt{a^2 + b^2}$$

Find a shorter side

Square, subtract, then square root.

$$a = \sqrt{c^2 - b^2}$$

Rounding

Give answers to 3 significant figures.

$$\sqrt{74} = 8.60$$

Examples 1 and 2 — Finding the Hypotenuse

EXAMPLES · 5 MIN

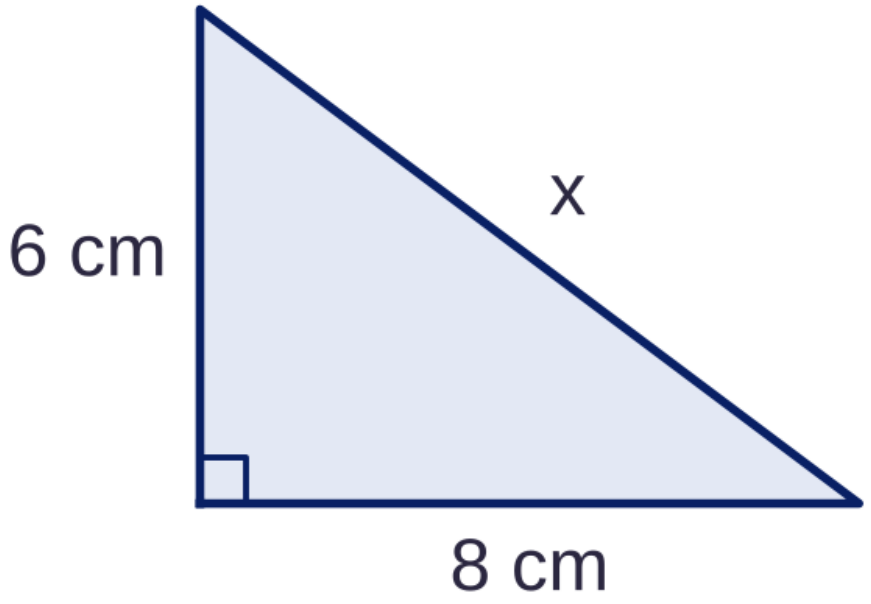
BRONZE

Watch and copy into your book

Hint $a^2 + b^2 = c^2$ c = hypotenuse

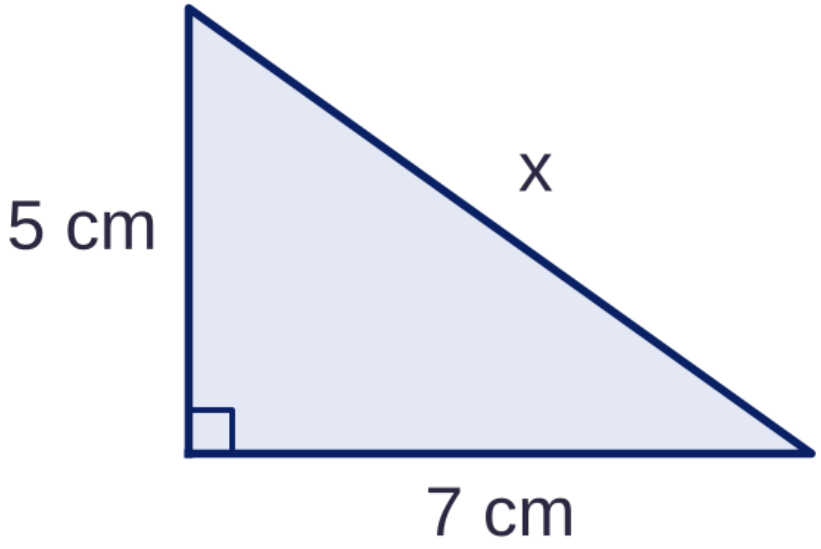
Example 1

Find the length x.



Example 2

Find x. Give your answer to 3 s.f.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$$x^2 = 6^2 + 8^2$$

$$x^2 = 100$$

$$x = \sqrt{100}$$

= 10 cm

Tip: the hypotenuse is opposite the right angle.

Example 2

$$x^2 = 5^2 + 7^2$$

$$x^2 = 74$$

$$x = \sqrt{74}$$

= 8.60 cm

Tip: square root at the end.

Task 1

PRACTICE · 9 MIN

BRONZE

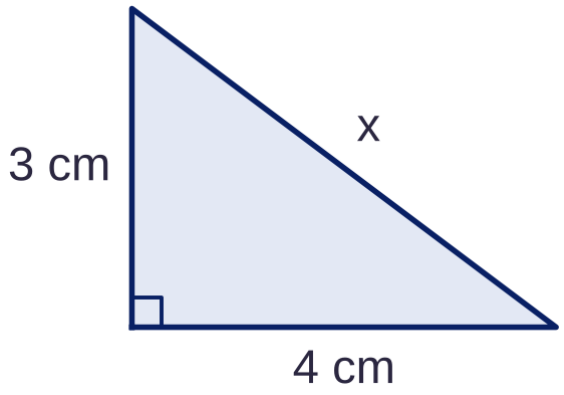
Find the length of the hypotenuse x (give answers to 3 s.f. where needed)

Hint

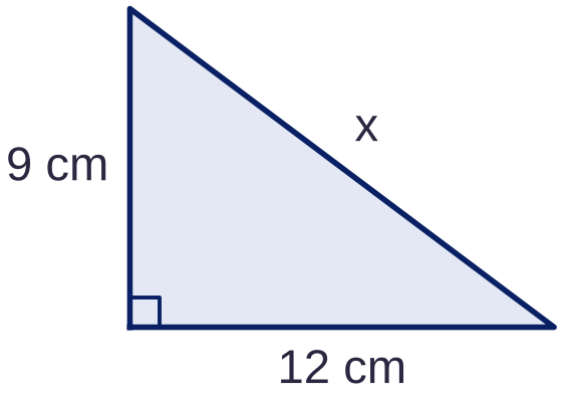
$a^2 + b^2 = c^2$

c = hypotenuse

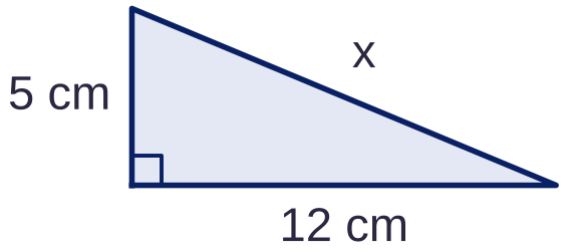
1



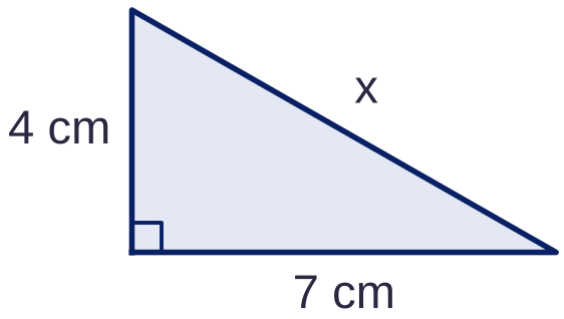
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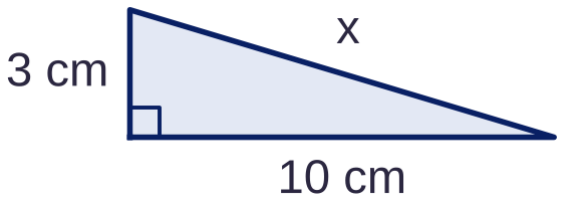
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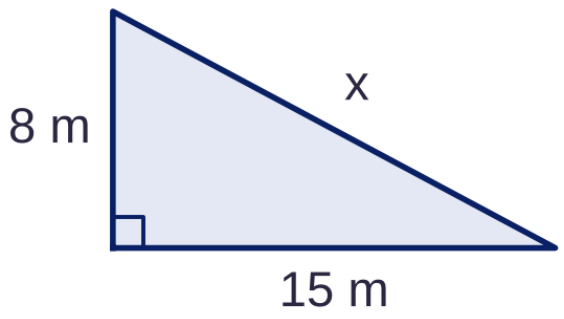
4



5



★



Task 1 — Answers

CORRECTIONS

BRONZE *Tick or correct in green pen*

1 $x^2 = 9 + 16 = 25, \quad x = \mathbf{5\ cm}$

2 $x^2 = 81 + 144 = 225, \quad x = \mathbf{15\ cm}$

3 $x^2 = 25 + 144 = 169, \quad x = \mathbf{13\ cm}$

4 $x^2 = 16 + 49 = 65, \quad x = \mathbf{8.06\ cm}$

5 $x^2 = 9 + 100 = 109, \quad x = \mathbf{10.4\ cm}$

★ $x^2 = 64 + 225 = 289, \quad x = \mathbf{17\ m}$

Examples 3 and 4 — Finding a Shorter Side

EXAMPLES · 5 MIN

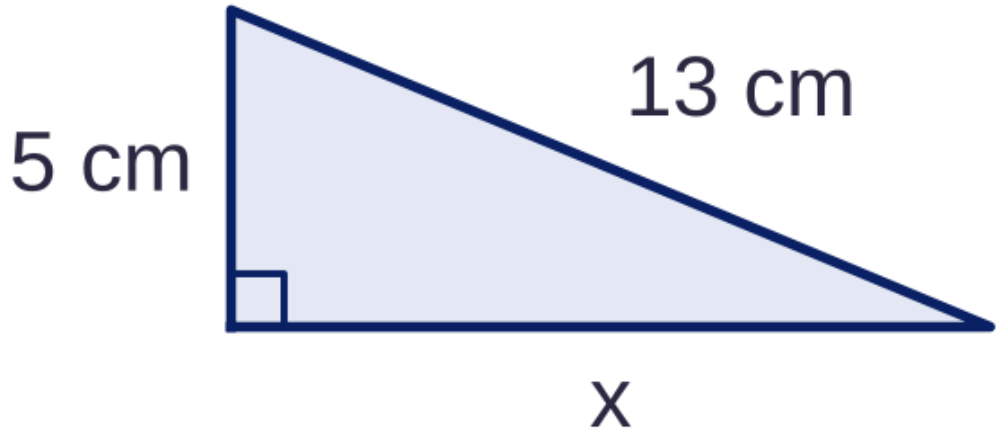
SILVER

Watch and copy into your book

Hint $a^2 + b^2 = c^2$ c = hypotenuse

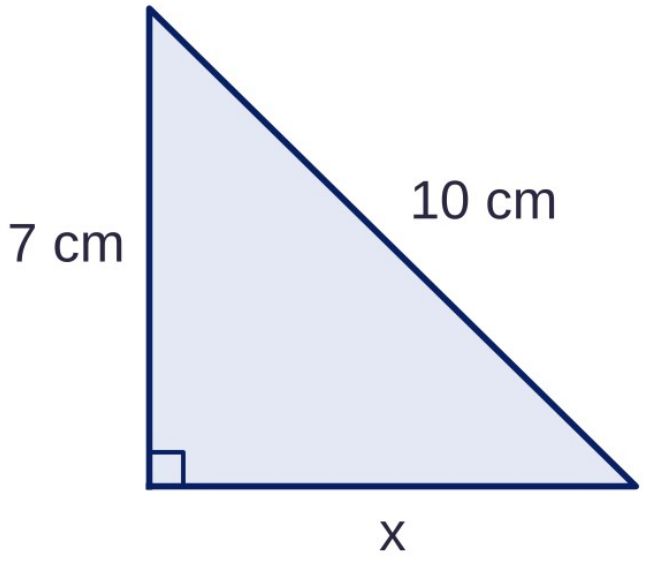
Example 3

Find the length x.



Example 4

Find x. Give your answer to 3 s.f.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$x^2 = 13^2 - 5^2$$

$$x^2 = 144$$

$$x = \sqrt{144}$$

= 12 cm

Tip: subtract when finding a shorter side.

Example 4

$$x^2 = 10^2 - 7^2$$

$$x^2 = 51$$

$$x = \sqrt{51}$$

= 7.14 cm

Tip: the hypotenuse squared comes first.

Task 2

PRACTICE · 9 MIN

SILVER

Find the shorter side (give answers to 3 s.f. where needed)

Hint $a^2 + b^2 = c^2$ c = hypotenuse

1. Hypotenuse 25 cm, one side 7 cm
2. Hypotenuse 17 cm, one side 8 cm
3. Hypotenuse 12 cm, one side 9 cm
4. Hypotenuse 9.5 m, one side 6 m
5. Hypotenuse 20 cm, one side 11 cm

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $x^2 = 625 - 49 = 576, \quad x = \mathbf{24\ cm}$

2. $x^2 = 289 - 64 = 225, \quad x = \mathbf{15\ cm}$

3. $x^2 = 144 - 81 = 63, \quad x = \mathbf{7.94\ cm}$

4. $x^2 = 90.25 - 36 = 54.25, \quad x = \mathbf{7.37\ m}$

5. $x^2 = 400 - 121 = 279, \quad x = \mathbf{16.7\ cm}$

Examples 5 and 6 — Pythagoras Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint $a^2 + b^2 = c^2$ c = hypotenuse

Example 5

(3 marks)

A ladder 6.5 m long leans against a wall.

The foot of the ladder is 2.5 m from the wall.

How high up the wall does it reach?

Example 6

(3 marks)

Find the distance between the points (1, 2) and (7, 10).

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$h^2 = 6.5^2 - 2.5^2$

$h^2 = 42.25 - 6.25$

$= 36$

$h = 6$ m

Tip: draw a sketch: the ladder is the hypotenuse.

Example 6

Across = 6

Up = 8

$d^2 = 6^2 + 8^2$

$= 100$

$d = 10$

Tip: make a right-angled triangle from the points.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint $a^2 + b^2 = c^2$ c = hypotenuse

1. A rectangle is 8 cm by 5 cm. Find the length of its diagonal. Give your answer to 3 s.f. (3 marks)

2. A ship sails 12 km north, then 9 km east. How far is it from its start? (3 marks)

3. An isosceles triangle has sides 10 cm, 10 cm and base 12 cm. Find its height. (3 marks)

4. Find the distance between (−2, 3) and (4, −5). (3 marks)

5. A triangle has sides 7 cm, 24 cm and 25 cm. Is it right-angled? Show your working. (2 marks)

Task 3 — Answers

CORRECTIONS

GOLD *Tick or correct in green pen*

1. $d^2 = 64 + 25 = 89$, $d =$ **9.43 cm**

2. $d^2 = 144 + 81 = 225$, $d =$ **15 km**

3. $h^2 = 100 - 36 = 64$, $h =$ **8 cm**

4. $d^2 = 6^2 + 8^2 = 100$, $d =$ **10**

5. $7^2 + 24^2 = 49 + 576 = 625 = 25^2$, **Yes**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE The two short sides are 12 cm and 16 cm. Find the hypotenuse.

SILVER The hypotenuse is 15 cm and one side is 9 cm. Find the other side.

GOLD Find the distance between (2, 1) and (5, 5).

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $x^2 = 144 + 256 = 400, \quad x = \mathbf{20\ cm}$

SILVER $x^2 = 225 - 81 = 144, \quad x = \mathbf{12\ cm}$

GOLD $d^2 = 3^2 + 4^2 = 25, \quad d = \mathbf{5}$

Exam Practice — Finding the Hypotenuse

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

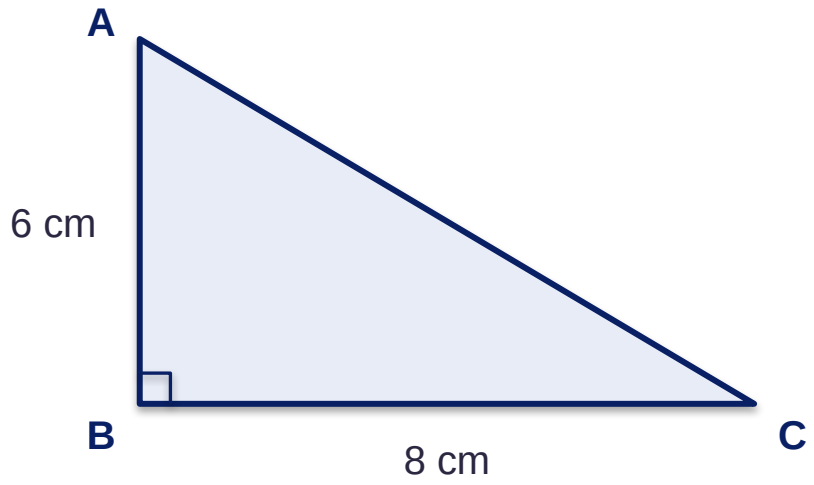
Hint $a^2 + b^2 = c^2$ c = hypotenuse

Question 1

(2 marks)

Work out the length of AC.

Diagram NOT accurately drawn



..... cm

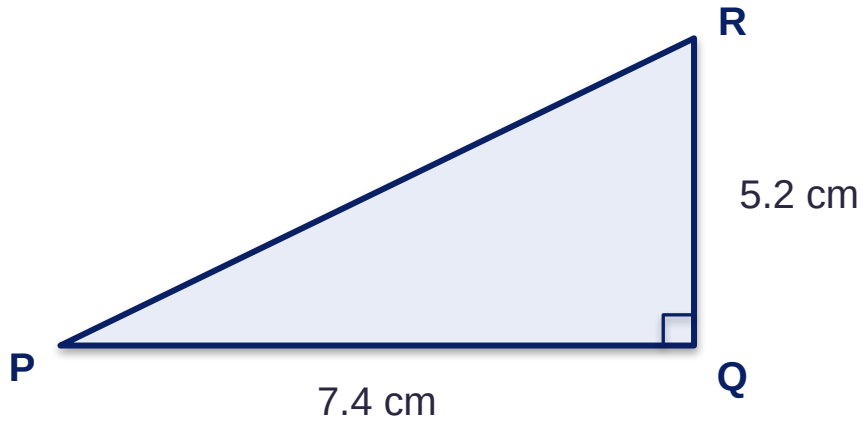
Question 2

(3 marks)

Work out the length of PR.

Give your answer correct to 1 decimal place.

Diagram NOT accurately drawn



..... cm

Exam Practice — Finding a Shorter Side

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

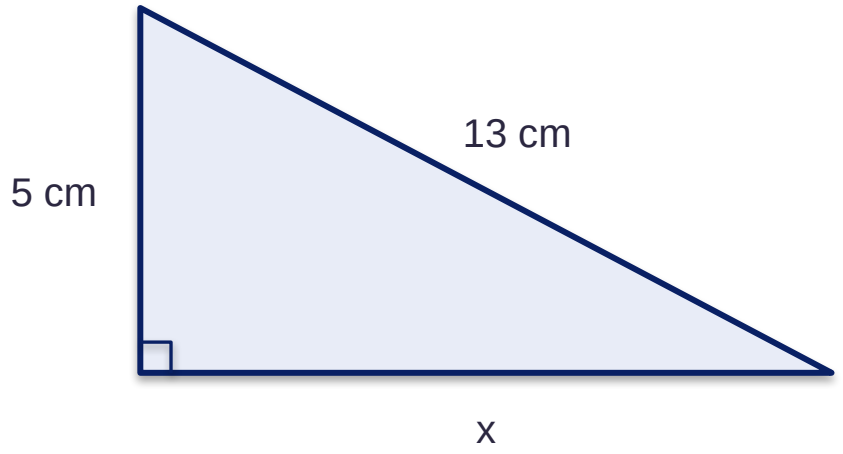
Hint $a^2 + b^2 = c^2$ c = hypotenuse

Question 3

(3 marks)

Work out the length of x.

Diagram NOT accurately drawn



..... cm

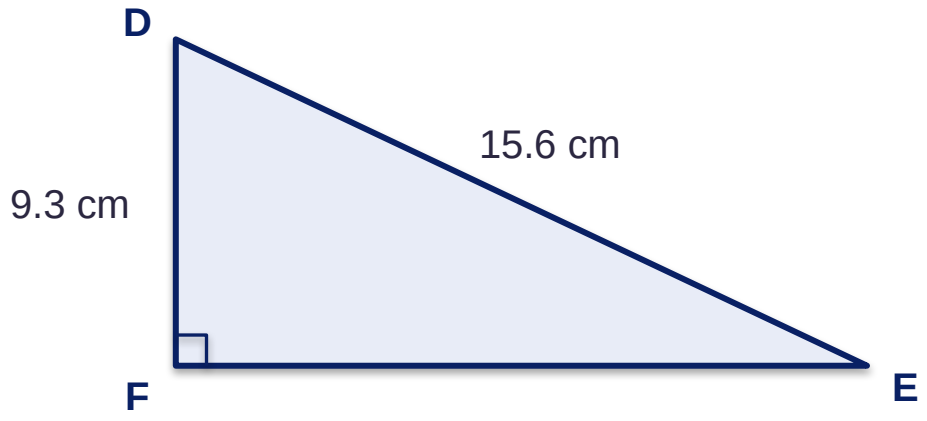
Question 4

(3 marks)

Work out the length of EF.

Give your answer correct to 3 significant figures.

Diagram NOT accurately drawn



..... cm

Exam Practice — Pythagoras in Context

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $a^2 + b^2 = c^2$ c = hypotenuse

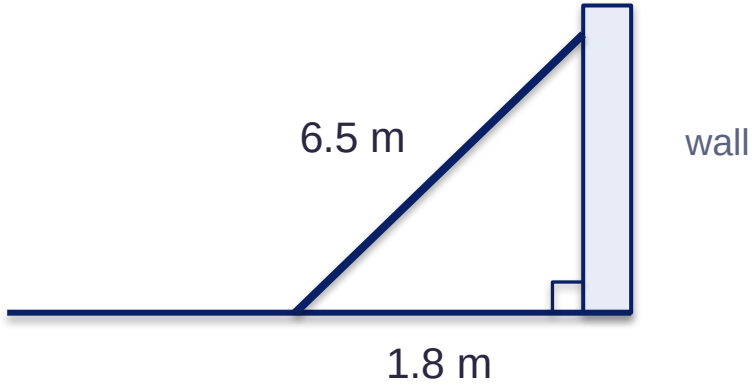
Question 5

(3 marks)

A 6.5 m ladder leans against a vertical wall. Its foot is 1.8 m from the wall.

How far up the wall does the ladder reach? Give your answer to 3 s.f.

Diagram NOT accurately drawn



..... m

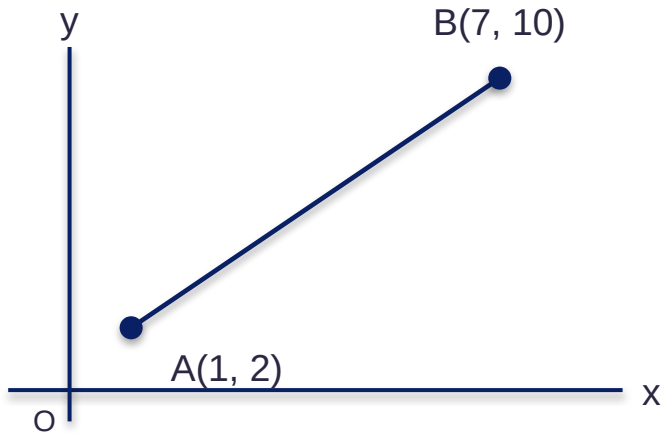
Question 6

(3 marks)

A is the point (1, 2). B is the point (7, 10).

Work out the length of the line AB.

Diagram NOT accurately drawn



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Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$6^2 + 8^2 = 100$ — M1

$\sqrt{100} = 10$ cm — A1

Tip: hypotenuse^2 = a^2 + b^2.

Question 2

$5.2^2 + 7.4^2 = 81.8$ — M1

$\sqrt{81.8} = 9.044\ldots$ — M1

9.0 cm — A1

Tip: write the full calculator value, then round.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$13^2 - 5^2 = 144$ — M1

$\sqrt{144} = 12$ — M1

x = 12 cm — A1

Tip: shorter side → SUBTRACT the squares.

Question 4

$15.6^2 - 9.3^2 = 156.87$ — M1

$\sqrt{156.87} = 12.524\ldots$ — M1

12.5 cm — A1

Tip: the hypotenuse is opposite the right angle.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$6.5^2 - 1.8^2 = 39.01$ — M1

$\sqrt{39.01} = 6.2458\ldots$ — M1

6.25 m — A1

Tip: the ladder is the hypotenuse.

Question 6

$\Delta x = 6,\ \Delta y = 8$ — M1

$6^2 + 8^2 = 100$ — M1

AB = 10 — A1

Tip: sketch a right-angled triangle under AB.